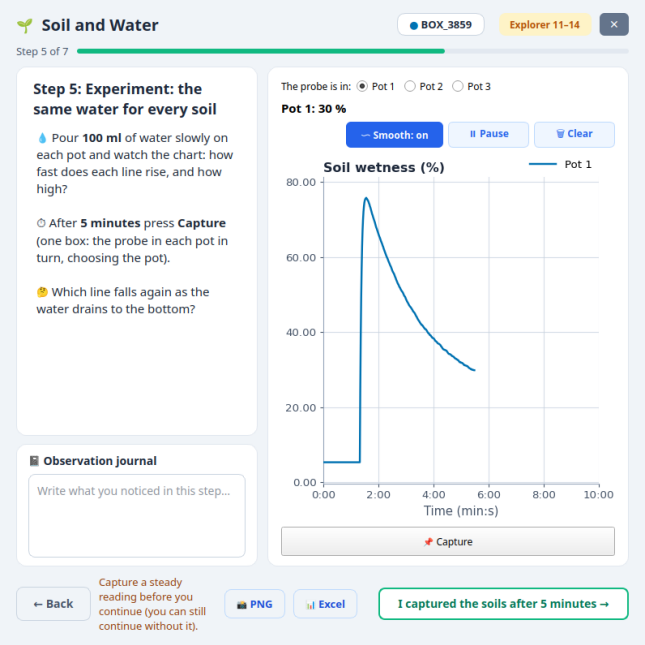
Soil and Water

Three soils, the same glass of water — and a probe the pupils calibrate

Level: Explorer and Scientist — ages 11–19 (Romanian school years V–XII) • **Time in class:** one 50-minute lesson; an optional extension over a few days • **Sensors:** the STEMMA soil probe (wetness); the light sensor, optional



The activity screen at the water step. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer and Scientist — ages 11–19 (Romanian school years V–XII). In year V (geography: soil) and year VI (biology: plants and water) pupils compare soils and read graphs; at upper secondary level they add the discussion of porosity, infiltration and floods (geography, year IX) and of calibrating an instrument.

Assumed knowledge

Reading a graph over time and the idea of a percentage. Nothing about calibration: the activity does it.

Time in class

One 50-minute lesson: calibration (10), the dry soils (10), the water and the 5-minute wait (15), the analysis (15). The extension over a few days costs no class time.

Teacher preparation

20 min — prepare three identical pots with drainage holes, filled with **sand**, **garden soil** and **clay** (or clayey soil), all left to **dry** for a few days and pressed down the same way. Stand the pots on trays. A measuring jug, a glass of water for the calibration and paper towels.

Sensor settling time

The probe gives one value a second. After you push it into the soil, wait **ten seconds** for the value to settle; after the water is poured, the line climbs within seconds and falls over minutes.

Sensors and boxes The **STEMMA soil probe** (Adafruit), read by firmware 2.5.0 or newer: a raw capacitance value, about 200 in air and up to about 2000 in water, and different from one probe to the next. It also reports the temperature of its own chip, which the lesson does **not** use: it follows the soil very slowly and is not a soil thermometer. The lesson runs with **a single box**, moved from pot to pot, or with **three boxes**, one in each pot, on the same chart. Classroom boxes, which have no probe, cannot run the lesson.

Three routes

Short One box, 50 min: the probe is calibrated once, then goes through the three pots at every measurement. The pots are watered one after another, and their 5-minute waits overlap.

Full Three boxes, 50 min: one probe in each pot, each one calibrated; the three lines appear side by side, and the comparison is at its clearest.

Project Extension over a few days: one probe in a pot in the sun and one in the shade, with logging switched on; a light sensor on the same box shows how much sun each one received. The data open from **Data & Export**.

The application starts in English. Buttons are named here exactly as they appear on the screen.

The lesson is hidden: switch it on once in Settings

The lesson needs the soil probe, which the usual boxes do not have, so the app does not show it by default. In **Settings**, under **Visible activities**, tick *Soil and Water*; the lesson appears at once on the **Activities** page. The tick stays on that computer.

1. Where this fits in the curriculum

- **Geography** — in year V, soil and its components; in year IX, water in the soil, infiltration and surface runoff: why the same rain floods one piece of ground and not another.
- **Biology** — in years VI–VII, plants and water: why the same plant dries out in sand and rots in clay; in year X, soil as a living environment.
- **Physics and mathematics** — a two-point calibration is a linear function: the value in air becomes 0%, the value in water 100%, and everything in between is shared out in proportion.

The core skill is **comparing three materials under equal conditions**: the same water, the same probe depth, the same waiting time.

2. Time plan

Stage	Time	What happens
Calibration	10 min	The probe in air, then in water; the app remembers the calibration for each box.
The dry soils	10 min	One reading captured for each pot.
The water	15 min	100 mL in each pot; the chart shows the rise and the drainage; after 5 minutes, one reading captured for each.
Analysis	15 min	The app's table: dry, highest, after 5 minutes, water held. The order of the soils and the explanation.

3. Materials

Item	Per group	Notes
MagicBox with the soil probe	1 or 3	With three boxes, one per pot.
Laptop with the app	1	The lesson has to be ticked once in **Settings** → **Visible activities**.
Three identical pots with holes, on trays	3	At least 10 cm deep, so the probe goes in up to its line.
Sand, garden soil, clay	one pot each	All dry, pressed down the same way.
Measuring jug	1	For the 100 mL of water in each pot.
Glass of water, paper towels	1	For the calibration and for wiping the probe between pots.

4. Scientific background for the teacher

What the probe measures and why we calibrate it

The probe does not count drops of water. It measures the **electrical capacitance** between two electrodes on its board: water has a permittivity about 80 times that of air, the soil's minerals about 4–8 times, so the more water there is around the probe, the higher the number. The number is **raw**: it depends on the probe, on how well it touches the soil, and on salt.

The two-point calibration. Pupils capture the value in air (a) and in water (w). For any reading x, the app computes

$$u = \frac{x - a}{w - a} \cdot 100\,\%,$$

limited to between 0 and 100 %. The result is not the soil's water content in the agronomic sense; it is a **relative wetness of the probe**, which we can compare between pots and between probes. If the two values are too close (the probe held in air twice, for example), the app refuses the calibration and asks for it to be done again.

Why soils hold water differently. Sand has large grains and large gaps: water gets in quickly and drains quickly. Clay has very small particles and very small gaps; water gets in with difficulty but is held tightly by the enormous surface of the particles. Garden soil also contains plant remains, which swell and hold water like a sponge.

4.1 What to expect

Soil	The chart after watering	After 5 minutes
Sand	climbs very high at once	falls quickly; holds little
Garden soil	climbs within a few seconds	stays high; holds well
Clay	climbs slowly, sometimes barely	water may pool on top; the probe sees little at first, then more and more

The exact values depend on how dry and how tightly pressed the soils are and on the depth of the probe. Watch the **order** of the soils, not a number.

5. Lesson flow

The in-app wizard has seven steps: the introduction, the calibration, the prediction, the dry soils, the water, the analysis and the conclusion.

1. **The introduction (3 min).** Rain on a field, on a garden and on a sandy path. The activity opens from the **Activities** page with **Start**.
2. **The calibration (10 min).** For each box, the probe held in the air, touching nothing: **In air**; then dipped in water up to the line on the probe: **In water**. Next to each box appears *Calibrated: air …, water …*. A box calibrated yesterday already shows this; calibrate again if the numbers look odd.

Technical note — the calibration, the pots and what is saved

Where the calibration is kept. In the app, for each box, on the computer where it was done. On another computer, calibrate again.
One box or three. With a single box, **The probe is in:** appears above the chart, with **Pot 1**, **2**, **3**; pupils choose the pot before each **Capture**. With two or three boxes, the first one connected is pot 1, the second pot 2 and so on, and a single **Capture** saves them all.
The chart. Soil wetness, in per cent, one line for each pot. During the water step the app also remembers each pot's highest value, for the table in the analysis.
What is not used. The temperature of the probe's chip (`SOIL_T` in the export) is not the temperature of the soil.
Exporting. The box's CSV log keeps the raw value `SOIL`; the percentages can be recomputed from it with the two calibration values shown on screen.

3. **The prediction (3 min).** *Which soil gets wet fastest? Which is still wet after five minutes?*
4. **The dry soils (10 min).** The probe in each pot, at the same depth; ten seconds; **Capture**.
5. **The water (15 min).** 100 mL poured slowly into each pot; watch the chart. After five minutes, **Capture**.
6. **Analysis and conclusion (15 min).** The app's table; the order of the soils; why.

The investigation, in one table

The question. Which soil holds water best?
The hypothesis. The order of the soils, written before the watering.
What I change. Only the soil.
What I measure. The relative wetness of the probe: dry, highest, after five minutes.
What I keep the same. The amount of water, how fast it is poured, the depth of the probe, how firmly the soil is pressed, the pots.
How many repeats. With three boxes, swap the probes between the pots and repeat: if the order stays, the probes were not what made the difference.
What makes a reading valid — and what makes it an artefact. The probe pushed in up to its line, with no air around it; do not pour the water straight onto the probe, which would then measure the water running down it, not the soil.

6. Analysis after the activity

6.1 Questions to ask afterwards

1. Which soil rose fastest? Which held the most after five minutes? Are they the same soil?
2. Why does the probe have to be calibrated? What would happen if we compared two uncalibrated probes?

3. Where would you plant a cactus, and where a geranium?
4. A town with a lot of asphalt and little soil: what happens to a heavy downpour?

6.2 Assessment ideas

- **Low stakes:** the pupil completes the table and puts the soils in order.
- **Standard:** the pupil explains the order by the size of the grains and of the gaps between them.
- **Extension:** the pupil explains the two-point calibration and works out the percentage for a raw value without help.

Common pitfalls and misconceptions

The probe at different depths. It measures a different layer of soil, and the comparison is no longer fair. The line on the probe is the mark.
Water poured onto the probe. The value jumps to that of plain water. Pour at the edge of the pot.
A calibration done with a wet probe. The "in air" value comes out too high. Wipe the probe first.
The probe's temperature taken for the soil's. It is the temperature of the probe's chip.
Clay that will not get wet. The water pools on top and the probe sees little: that is a result, not an error. Dry clay repels water at first.

Safety and classroom management

Water and electronics. Only the tip of the probe goes into the soil or the water; the box stays away from the trays, on a dry surface.
Hands. Garden soil may contain debris; gloves, or hand-washing after the lesson.
Glass. Plastic measuring jugs are safer than glass ones.

Adaptations for pupils with special educational needs

One pot per pupil. Each pupil looks after one pot: fills it, pours the water, announces the value.
No calculation. Pupils who do not yet work with percentages compare only the lines on the chart: which climbs first, which falls first.
Motor skills. A classmate can pour the water while the pupil times and reads the value.

If things do not go to plan

Only one box. The lesson works: the probe goes through the pots in turn. Water all three pots at almost the same time and measure them one after another after five minutes.
You have no clay. Use peat-based potting compost instead; the contrast with the sand stays clear.
The calibration is refused. The values in air and in water are too close: check that the probe is dry for the first and dipped up to its line for the second.
The box will not connect. Use the "What to expect" table as a dataset and discuss the order of the soils.

7. Further resources

- **FAO** — `https://www.fao.org` — search for *soil texture* and *soil water*: material on the world's soils and the water in them.

- **The National Research and Development Institute for Soil Science, Agrochemistry and Environment (ICPA)** — `https://www.icpa.ro` — the soils of Romania.
- **Adafruit** — `https://learn.adafruit.com` — search for *STEMMA soil sensor*, for how the probe works.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *which soil would you plant a thirsty plant in, and why?*

On the **common three-axis rubric**:

- **Measurement practice** — the pupil calibrates the probe, pushes it in to the same depth and pours the same water.
- **Data interpretation** — the pupil reads the order of the soils from the chart and from the table.
- **Scientific reasoning** — the pupil links the order to the size of the grains and states a limitation: the probe sees only a few centimetres.

The rubric itself, with descriptors for all levels, is defined in the *curriculum plan*.

9. Pupil sheet

The sheet is a separate document, in `docs/en/student_sheets`: the file `soil_and_water.pdf`. It has the usual layers: what I see on the screen, what I do on my own, the conclusion I write. The table has one row for each pot and a column each for dry, highest and after five minutes.